Balancing the system

The predator–prey cycles identified by Lotka and Volterra were focused on the interaction between single predator and prey species. Since their studies, the theory of feedback loops has developed to embrace entire ecosystems. Ecologists now think that negative feedback loops are of central importance for the functioning of all ecosystems, keeping every part of them naturally within the bounds of sustainability. Populations can never swell for long beyond the carrying capacity of the rest of the system to support them. Thus, negative feedback regulates an ecosystem and keeps it stable.

Positive feedback interferes with a balanced ecosystem. If there is a surplus of resources, or a lack of predators, a population can grow freely. A bigger population leads to more births, and so an acceleration of the growth in population.

Equally, positive feedback can result in an accelerated contraction of a population. If fish stocks decline in a lagoon, for instance, local people may resort to importing canned food. Pollution from the dumps where the cans are thrown away can seep into the lagoon, killing the fish—and encouraging the locals to import even more of the damaging cans. And yet, positive feedback loops can sometimes set off a chain of events that becomes a “virtuous” circle. For example, if shrubs are planted in unstable soil, their roots may stabilize the soil, allowing both the shrubs and soil to thrive.